AI TOOLS FOR RESEARCH

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Review topic: Comparing AI-Assisted and Manual Data Labelling Accuracy on Scientific Datasets

Prompt 01: Research Workspace

Research area and previous work

My literature review compares manual, AI-only and human-in-the-loop annotation of scientific data, covering text classification, biomedical screening, metadata extraction, image annotation, active learning and label quality. The review concludes that annotation quality depends on task structure, reference labels, model choice and verification practices [1].

Workspace and evidence tracking

The previous review and bibliography form the initial evidence base. I will maintain a research_decisions.md record with each decision, reason, supporting source and next action. Relevant statements are marked VERIFIED (checked directly), INFERRED (an interpretation or proposed test), or UNKNOWN (access or evidence not yet confirmed). A claim mentioned in the review is not automatically verified from its original publication.

Initial decisions

1. Continue within scientific data labelling to preserve the connection with the earlier review [1].

2. Prioritize selective verification and matched review effort as candidate questions arising from the review [1].

3. Verify the original papers, data terms, baselines, evaluation protocol and available compute before presenting an experiment as validated.

Prompt 02: Research Gaps

Evidence from the review

Manual annotation supports context-sensitive decisions but may face disagreement and effort costs [1, 2]. Automated annotation is useful on structured tasks but is inconsistent across domains [1, 3, 4]. Human verification can correct AI predictions, but reviewer behavior is difficult to isolate [1, 5]. Existing benchmark labels may themselves be incorrect [1, 6, 7]. Active-learning literature motivates prioritizing cases rather than examining them indiscriminately [8].

Five computationally testable gaps

1. Selective verification: The review identifies limited evaluation of integrated active learning and verification (Sections 3.4, 5.2 and 6) [1]. Test whether uncertain predictions contain more errors than randomly chosen predictions under the same review budget.

2. Accuracy at matched review effort: Cost and quality are often reported separately (Sections 5.2 and 6) [1]. Compare accuracy and macro-F1 at fixed numbers of reviewed cases; count is a proxy, not measured human time.

3. Imperfect reference labels: Benchmark label error can affect comparisons (Sections 2.3, 5.1 and 6) [1, 6, 7]. Study performance sensitivity under controlled synthetic label noise without contaminating protected test references.

4. Scientific-domain transfer: Evidence is concentrated in biomedical and English-language work (Sections 5.3 and 6) [1]. Compare the same labeling protocol on two compatible scientific text datasets, if accessible.

5. Version reproducibility: Changing commercial models complicate reproducibility (Sections 5.5 and 6) [1]. Compare predictions from documented model versions or fixed configurations.

Top three gaps by significance and feasibility

1. Selective verification: direct relevance to the review and manageable computational testing with a fixed classifier and reference-label simulation.

2. Matched review effort: a fair controlled comparison is practical when effort is expressed as the number of items reviewed; actual labor cost remains unmeasured.

3. Reference-label noise: methodologically important, but requires safeguards around benchmark evaluation and a carefully controlled noise model.

These are candidate gaps identified from the review; claims of novelty and the status of individual original-study limitations require independent source checking.

Prompt 03: Map Literature to an Executable Experiment

Published methods and relationship to the proposed study

1. Automated text annotation: Gilardi et al. [3] compare automated labels with crowd workers on text-annotation tasks. The setting is not SciCite citation intent; direct replication would require confirming the exact data, prompts and model version.

2. Confident learning: Northcutt, Jiang and Chuang [7] study identifying label problems using model predictions. This motivates quality auditing, but reproducing their published experiments would require separate protocol verification.

3. Uncertainty sampling: Settles [8] surveys choosing informative examples for annotation. Our proposed fixed-model correction queue is distinct from active learning because no model retraining follows the corrections.

Contribution pathways

A. Reproduction with extension: Reproduce a published annotation method and vary review conditions; potentially greater model/data access burden.
B. Controlled comparison (selected): Compare random and low-margin selection under fixed review budgets, using lightweight TF-IDF classifiers as baselines.
C. Targeted modification: Alter the uncertainty score and perform an ablation; deferred until baseline validation.

Literature-to-experiment chain: previous review gap [1] → uncertainty prioritization [8] → SciCite citation-intent data [9] → TF-IDF Logistic Regression and linear SVM → matched-budget correction simulation. Code/weights access and the specific reproducibility claims must be verified against original sources; the proposed lightweight classifiers are not reproductions of SciCite's neural model.

Prompt 04: Feasibility, Novelty and Resource Access

Feasibility and access

The proposed SciCite task [9] assigns citation contexts to background, method or result. The public dataset source and its JSONL schema have been inspected during subsequent notebook work, but public access alone does not settle raw-data redistribution rights. TF-IDF and linear classifiers can run on Google Colab CPU; a GPU is not essential for the proposed minimum baseline. The actual environment and versions must be documented when run.

Novelty and evidence checks

Search for neighboring work using: “SciCite uncertainty error detection”; “citation intent active learning”; “label verification review budget”; and “oracle correction uncertainty sampling”. Compare each relevant original method, its verified contribution, the reported limitation, and the specific difference in this study. This is a controlled course-level comparison; no new selection algorithm or publication-level novelty is claimed.

Risks and decision

Risks: dataset permissions, shared citing-paper identifiers across splits, imperfect reference labels, and a simulated oracle that cannot measure real reviewer performance. **Go/no-go:** proceed with the comparison if data access and baseline protocol are verified. If SciCite becomes unavailable, select another legitimately accessible scientific-text corpus and record the change before execution.

Prompt 05: Research Question and Success Criteria

Problem: Given limited capacity to review automated scientific citation labels, determine whether uncertainty-based prioritization identifies more incorrect predictions than random selection at an equal number of reviews.

RQ1: At fixed review budgets, does low-margin SVM selection recover more errors and yield higher final annotation accuracy than random selection?

RQ2: Does the direction of the comparison remain on held-out data and after filtering test examples that share citing-paper IDs with the training set?

Inputs and comparators: SciCite citation contexts and reference labels; no correction, random selection and uncertainty-based selection. The selected examples are corrected by replacing predictions with reference labels for a perfect-oracle simulation.

Primary outcome: accuracy (higher is better). **Secondary outcomes:** macro-F1 and the number of errors recovered. The review budget, dataset partition, model parameters and selection-count protocol are held constant between strategies. A smaller or negative improvement for uncertainty selection would contradict the expected advantage.

Interpretation boundary: This experiment tests prioritization under perfect simulated correction. It cannot establish actual human reviewer speed, cost, verification error or automation bias. Baseline and test results are documented in later phases, not claimed as work performed under these planning prompts.

References

[1] Nikam, U. Comparing AI-Assisted and Manual Data Labelling Accuracy on Scientific Datasets. Earlier literature review, Sections 2–6.

[2] Al-Athba, A. A., and Zaghouani, W. (2026). Cohesion-6K: An Arabic dataset for analyzing social cohesion and conflict in online discourse. arXiv:2605.22447.

[3] Gilardi, F., Alizadeh, M., and Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. PNAS, 120(30).

[4] Turner, M. D., et al. (2025). Large language models can extract metadata for annotation of human neuroimaging publications. Frontiers in Neuroinformatics, 19.

[5] Chen, H., et al. (2025). A human-LLM collaborative annotation approach for screening articles on precision oncology randomized controlled trials. BMC Medical Research Methodology, 25.

[6] Northcutt, C. G., Athalye, A., and Mueller, J. (2021). Pervasive label errors in test sets destabilize machine learning benchmarks. NeurIPS.

[7] Northcutt, C. G., Jiang, L., and Chuang, I. L. (2021). Confident learning: Estimating uncertainty in dataset labels. JAIR, 70.

[8] Settles, B. (2009). Active learning literature survey. University of Wisconsin–Madison Technical Report 1648.

[9] Cohan, A., Ammar, W., van Zuylen, M., and Cady, F. (2019). Structural Scaffolds for Citation Intent Classification in Scientific Publications. NAACL-HLT.